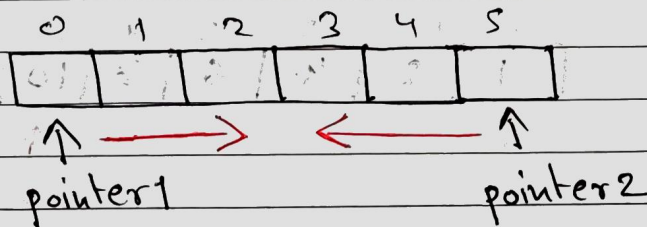


Two Pointer Technique

It is a common algorithmic technique in which two different pointers are maintained i.e. i & j to traverse the given data structure (Usually Arrays and Linked List).

Basic Idea:-

- > Two different pointers pointing at different or same memory location either in
- > They move in different direction or at a different speed.



Example to understand two pointer

Q. Find two Number that sum to a Target number.

[1, 2, 4, 6, 8, 10]

Target = 10

Sorted Array

Solution:-

To solve this we will use pointers
i & j

i, will point at index 0

j, will point at the end [idx 5]

0	1	2	3	4	5
1	2	4	6	8	10

↑
i
↑
j

Condition:

- i, will move rightwards
if $i + j < \text{target}$
- j, will move leftward
if $i + j > \text{target}$

[1 2 4 6 8 10]

↑
i

↑
j

Target = 10

iteration 1

i = 1, j = 10;

*

loop will stop if
i & j cross each other
(i > j)

i + j = 11; greater than target

j++;

[1 2 4 6 8 10]

↑
i

↑
j

iteration 2

i = 1, j = 8;

i + j = 9; lesser than target

i++

[1 2 4 6 8 10]

↑
i

↑
j

iteration 3

$i = 2, j = 8;$

$i + j = 10;$ equals to target

return $[i, j]$

* loop will stop

- How to know where two pointer will be used or may work?

Solution :-

Look for these signs.

- 1. Question must be of array or linked list
- 2. If the given data is sorted or it can be sort
- 3. Merge If question say, Merge / Remove Duplicate / Rearrange
- 4. If say detect cycle in Linked List
- 5. find pair, or find more than one Ans

Flow Chart
to Guess this
pattern

Array / Linked list

Sorted / sort

Merge / Remove Duplicate /
Rearrange

Detect Cycle
in linked list

Pair / Triplet
Quadrant